

Application of a Numerical Model for the Multicomponent Alloy Solidification to a Phase Change Material Applied in Latent Heat Thermal Energy Storage

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The search for environmentally friendly energy management systems has become fundamental regarding the growing use of fossil fuel by the energy sector and its impacts on the environment. In this sense, latent heat thermal energy storage (LHTES) is a promising technology for recovering, storing, and subsequently using the heat generated from renewable sources, which in turn can make industrial facilities more efficient and reduce CO₂ emissions. The proposed numerical model aims to provide auxiliary tools for the design of thermal reservoirs by considering the kinetics and controlling parameters of phase transformations to better equate supply and demand. A medium temperature PCM ~900K and a previously developed numerical model are carried out to simulate the solidification process of Al-Cu-Mg-Fe alloys. The numerical model is coupled with theoretical SDAS predictions, enabling this model to be applied to investigate different heat extraction scenarios, considering a wide range of Biot and Fourier numbers.

Keywords: *Thermal energy storage design, Phase-change material, LHTES, Non-Equilibrium thermodynamics, Numerical Model.*

1 Introduction

Global warming mitigation demands decreasing GHG emissions and broad use of renewable energy resources¹. The design of latent heat thermal energy storage reservoirs to minimise the mismatch between supply and demand is necessary for centralized and distributed electrical power generation systems²⁻⁴. Latent heat thermal energy storage (LHTES) stores heat 5 to 14 times more than sensible heat thermal energy storage (SHTES)⁵⁻⁷. The thermal storage reservoirs use a phase-change material (PCM) to store energy in latent heat by absorbing high energy during melting and releasing it during solidification. As a result of the varying solar generation potential attributed to seasonality, weather conditions, and climate change, an intermediate medium is employed to transfer the thermal energy to the working fluid of low, medium and high melting/liquidus temperature, thereby facilitating thermal energy storage^{8,9}. Concerning the conversion of the storage thermal energy to electricity, the power block, it can be through thermal electric generator (TEG) and steam Rankine cycle. The maximum theoretical efficiency of TEG is 33% for high temperature around 1400 K^{10,11}, against ~64% of a combined steam Rankine cycle, which is the power block configuration most employed and commercially available¹².

In this paper, a numerical model previously applied to the study of solidification kinetics, solute segregation of multicomponent alloys¹³⁻¹⁶ will be used to predict the kinetics of solidification of a medium-temperature PCM and to elaborate graphical solution of the thermal variables of solidification for a wide range of Biot and Fourier numbers.

2. Mathematical Formulation and Numerical Method

The numerical model adapted to this work for the simulation of the solidification process of pure metals and eutectic alloys is the same proposed by Swaminathan and Voller¹³, modified and applied to problems of inverse macrosegregation of multicomponent systems^{14,16}.

The numerical model encloses the following features:

1. It can deal with the coupling of temperature and composition fields and multiple reactions during solidification of multicomponent alloys, such as primary phase transformation, phase precipitations, eutectic and peritectic transformations;
2. It considers the effect of the difference in specific mass between the liquid and other primary and secondary phases during the solidification in the form of a shrinkage-induced flow;
3. A temperature rescaling is used to deal with the solidification of pure eutectic materials;

2.1. Model assumptions

The numerical model for unsteady solidification of pure metals/compounds and multicomponent alloys is based on the following assumptions:

4. The domain is finite and along the axis z , defined by $0 \leq z \leq z_b$ according to Figure 1. The boundary conditions at $z = 0$ are of third kind for heat flux and second kind for mass, and at the boundary $z = z_b$ no heat and mass fluxes are allowed;

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5. The phase change material to be solidified is assumed as a pure metal or a multicomponent alloy;
6. The solid phase is stationary, i.e., once formed it has a velocity equal to zero;

2.2. Governing equations

The governing equations for the unsteady solidification of multicomponent alloys for general purpose must consider mass, energy and chemical species for general application,

- Mass

$$\frac{\partial}{\partial t} + \nabla \cdot (\mathbf{u}) = 0 \quad (1)$$

- Energy

$$\frac{\partial \rho C_P T}{\partial t} + \nabla \cdot (\rho C_P \mathbf{u} T) = \nabla \cdot (k \nabla T) - \rho_S \Delta H \frac{\partial g}{\partial t} \quad (2)$$

- Chemical Species for Component i

$$\frac{\partial \rho C_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} C_i) = \nabla \cdot (D_i \nabla C_i) \quad (3)$$

The velocity field in the liquid can be updated by the value of the liquid volume fraction, g_l as $\mathbf{u} = g_l \mathbf{u}_l$.

- Boundary and initial conditions:

$$\text{At } z = 0,$$

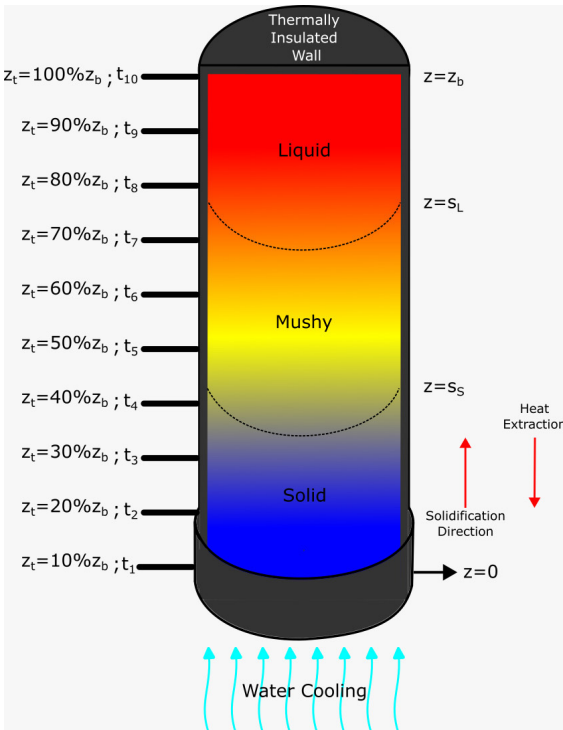


Figure 1. Schematic representation of transient upward solidification cooled from the bottom.

$$u = 0, -k \frac{\partial T}{\partial z} = h_g (T|_{z=0} - T_\infty) \text{ and } \frac{\partial C_i}{\partial z} \Big|_{z=0} = 0 \quad (4)$$

$$\text{At } z = z_b,$$

$$u \rightarrow u_\infty, \frac{\partial T}{\partial z} \Big|_{z=z_b} = 0 \text{ and } \frac{\partial C_i}{\partial z} \Big|_{z=z_b} = 0 \quad (5)$$

$$\text{At } t = 0, \text{ and } 0 \leq z \leq z_b,$$

$$u = 0, T(z) = -az^2 + bz + c, \text{ and } C_i = C_{0i} \quad (6)$$

The macroscopic concentration density field depends on the local concentration and the relationship is made by applying a microscale model^{13,17,18} as the following

$$\begin{aligned} & [\rho C_{L,i}] - [\rho C_{L,i}]_P^{old} + \\ & [C_{L,i}]_P = \frac{[\rho_{L,i} g_P^{old} + \beta_i \rho_{S,i} (1 - g_P^{old}) k_{0,i}] [C_{L,i}]_P^{old}}{\rho_{L,i} g_P^{old} + \beta_i \rho_{S,i} (1 - g_P^{old}) k_{0,i} + (1 - \beta_i) \rho_{S,i} k_{0,i} (g_P^{old} - g_P^{n+1})} \end{aligned} \quad (7)$$

where the parameter β_i is the solute redistribution parameter, defined such that $0 \leq \beta_i \leq 1$. The back-diffusion parameter β can be reckoned via microsegregation model^{19,20}.

$$\beta_i = \frac{\alpha_i \gamma_i}{\alpha_i + \gamma_i f \left(\frac{df}{d\tau} \right)} \quad (8)$$

and

$$\gamma_i = \frac{A \alpha_i k_i}{A \alpha_i k_i + 1} \quad (9)$$

and α_i is given by,

$$\alpha_i = \frac{4 D_i t_f}{SDAS^2} \quad (10)$$

where τ is the dimensionless time, f is the solid fraction, t_f is the solidification time for a Representative Elementary Volume¹⁹, D_i is the diffusion coefficient of solute i and $SDAS$ is the secondary arm spacing. A detailed description of $SDAS$ and its relationship to the Gibbs-Thomson coefficient can be found in^{19,20}. According to the authors $SDAS$ can be calculated in terms of Gibbs-Thomson coefficient^{20,21} given by,

$$SDAS = 5.5 (M \cdot t_{SL})^{\frac{1}{3}} \quad (11)$$

and,

$$M = \frac{-\Gamma \sum_{j=1}^n w_j \beta_j}{\sum_{j=1}^n m_j \left[1 - \frac{k_j}{k_j + (1-k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right] (c_{f,j} - c_{0,j}) / D_j} \quad (12)$$

$$\ln \left[\frac{\sum_{j=1}^n m_j \left[1 - \frac{k_j}{k_j + (1-k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right] c_{f,j} / D_j}{\sum_{j=1}^n m_j \left[1 - \frac{k_j}{k_j + (1-k_j) \exp\left(\frac{-v \delta_j}{D_j}\right)} \right] c_{0,j} / D_j} \right]$$

and the general Gibbs-Thomson coefficient Γ is expressed in terms of the Thermal Field Tensor Γ ²⁰⁻²⁵ given as

$$\Gamma = A \cdot \nabla T = A \cdot \nabla \left[\frac{\partial T}{\partial E} E \right] = A \cdot \left[\nabla \left(\frac{\partial T}{\partial E} \right) E + \frac{\partial T}{\partial E} \nabla E \right] \quad (13)$$

where the thermal field Γ ²¹⁻²⁵ provides

$$T = \frac{\partial T}{\partial E} E = \frac{\partial T}{\partial U} U + \frac{\partial T}{\partial K} K + \frac{\partial T}{\partial P} P + \frac{\partial T}{\partial W} W + \dots + \frac{\partial T}{\partial \Sigma_{other}} \Sigma_{other} \quad (14a)$$

and

$$T = \frac{\partial T}{\partial E} E = T_F - \sum_{j=1}^n \frac{\partial T}{\partial n_i} n_i + \frac{\partial T}{\partial \sigma} \sigma + \dots + \frac{\partial T}{\partial \Sigma_{other}} \Sigma_{other} \quad (14b)$$

denoting the thermal field gradient in terms of the First Law of Thermodynamics normal to a new created and deformed surface area.

The thermal field tensor Γ yields the relationship between nucleation and thermal gradient T . In this way, by considering second order nucleation formulation the following expression^{21,22,25} be derived,

$$\Gamma = -\frac{3}{4} \left[\frac{\left(\Delta S_V \Delta T + \frac{\partial \gamma_{SL}}{\partial r} \right) + \gamma_{SL} \frac{1}{f(\theta)} \frac{\partial f(\theta)}{\partial r}}{\frac{1}{\Delta T f(\theta)} \frac{\partial (\Delta S_V \Delta T f(\theta))}{\partial r}} \right] = \quad (15)$$

$$A \cdot \nabla T = A \cdot \left[\nabla \left(\frac{\partial T}{\partial E} \right) E + \frac{\partial T}{\partial E} \nabla E \right]$$

The solution of Equations 15 depends on calculation of surface energy γ_{SL} as a function of surface stress tensor δ and the vector radius r_C . While the thermal field T can be related to the stress tensor σ

By comparing the variation of excess of internal energy with the differentiated form of superficial internal energy and correlating with the Gibbs-Duhem equation^{26,27},

$$A d\gamma = -S^{surf} dT + N_j^{surf} d\mu_j + A_0 \sum_{\alpha\beta} (\Sigma_{\alpha\beta} - \gamma \delta_{\alpha\beta}) d\varepsilon_{\alpha\beta} \quad (16)$$

The surface stress and surface energy relationship, the so-called Shuttleworth equation²⁸, can be derived from Equation 17 by assuming no change in heat and in the number of particles on the surface. For anisotropic solids,

$$\Sigma_{\alpha\beta} = \gamma \delta_{\alpha\beta} + \frac{\partial \gamma}{\partial \varepsilon_{\alpha\beta}} \quad (17)$$

Considering the classical bulk and surface stress tensor Σ as derived by Gurtin and Murdoch²⁹ regarding the material surface equations can be given by

$$\text{div}_s \Sigma = \mathbf{T} \mathbf{n} + \rho_0 \ddot{\mathbf{u}} \quad (17a)$$

$$\Sigma = \sigma \mathbf{I} + 2(\mu_0 - \sigma) \mathbf{I} \mathbf{E} + (\lambda_0 - \sigma) (\text{tr} \mathbf{E}) \mathbf{I} + \sigma \nabla_s \mathbf{u} \text{ on surface } s \quad (17b)$$

$$\mathbf{E} = \frac{1}{2} (\mathbf{D} \mathbf{u} + \mathbf{D} \mathbf{u}^T) \quad (17c)$$

where $\mathbf{T}$, $\mathbf{E}$ and $\mathbf{u}$ are the stress, the strain and displacement fields, σ and ρ_0 are the residual surface tension and the surface mass density, and finally, λ_0 and μ_0 are the Lamé moduli.

A detailed solution scheme of numerical solution for transient solidification of multicomponent alloys can be found in¹⁴⁻¹⁸. Surface energy calculations are discussed in^{22,23,25}.

3. Results and Discussion

The finite volume solution for the system of PDE equations, as given by Equations 1-7, is described in detail in the literature¹³⁻¹⁶. The alloy composition for this study is Al-3wt%Cu-0.5wt%Mg-0.02wt%Fe. The macroscopic density fields are related to the microscale using Equation 7, in which the back-diffusion parameter β_i for each solute i must be determined for the local kinetics. A microsegregation model is then invoked^{19,20}, Equations 8-10 together with Equations 11 and 12 for the secondary arm spacing²⁰. To solve the SDAS, the Gibbs-Thomson coefficient Γ is not known previously, as it is a function of the local thermal gradient ∇T . The Gibbs-Thomson coefficient Γ is derived in terms of the first and second laws of Thermodynamics to predict the new surface area A created and deformed for non-equilibrium conditions, i.e. Equations 13-17, which Ferreira generalized Γ in the form of a thermal field tensor ∇T ^{22,25}. A relationship between the surface stress Σ and the surface energy γ_{SL} can be deduced from the surface internal energy by comparing it with the Gibbs-Duhem^{26,27} and Shuttleworth²⁸ equations. The surface energy γ_{SL} is a function of the surface stress Σ , the nucleation radius r_C and the angle θ .

The thermodynamic and kinetic data used in this work are provided in Table 1. The calculated undercooling ΔT , homogeneous $r_{C,Hom}$ and heterogeneous $r_{C,Het}$ nucleation radii, nucleation angle θ , $\frac{\partial f(\theta)}{\partial r}$, surface tension σ_{SL} , surface stress Σ , surface energy γ_{SL} , $\frac{\partial \gamma_{SL}}{\partial r}$ and Gibbs-Thomson

Table 1. Thermophysical properties of investigated alloy^{20,25}.

Properties	Symbol	Units	Al-3wt%Cu-0.5wt%Mg-0.2wt%Fe
Liquidus temperature	T_L	K	922.61
Solidus temperature	T_{SOL}	K	855.00
Liquidus slope	m_L^{Mg}	$K (wt\%Mg)$	-5.100
	m_L^{Cu}	$K (wt\%Cu)$	-3.641
	m_L^{Fe}	$K (wt\%Fe)$	-4.200
Partition Coefficient	k_0^{Mg}	-	0.36
	k_0^{Cu}	-	0.1308
	k_0^{Fe}	-	0.03
Activation energy (Al liquid)	$Q_d^{Mg}{}_L$	$kJ\ mol^{-1}$	71.6
	$Q_d^{Cu}{}_L$	$kJ\ mol^{-1}$	24
	$Q_d^{Fe}{}_L$	$kJ\ mol^{-1}$	35
Diffusion constant (Al liquid)	$D_0^{Mg}{}_L$	$m^2\ s^{-1}$	9.9×10^{-5}
	$D_0^{Cu}{}_L$	$m^2\ s^{-1}$	1.06×10^{-7}
	$D_0^{Fe}{}_L$	$m^2\ s^{-1}$	2.34×10^{-7}
Activation energy (Al FCC)	Q_d^{Mg}	$kJ\ mol^{-1}$	115.0
	Q_d^{Cu}	$kJ\ mol^{-1}$	136.1
	Q_d^{Fe}	$kJ\ mol^{-1}$	183.4
Diffusion constant (Al FCC)	D_0^{Mg}	$m^2\ s^{-1}$	6.23×10^{-6}
	D_0^{Cu}	$m^2\ s^{-1}$	6.50×10^{-5}
	D_0^{Fe}	$m^2\ s^{-1}$	5.3×10^{-3}
Latent heat (FCC _ Al)	ΔH_{FCC_Al}	$J\ kg^{-1}$	293900
Mean Thermal Gradient (Liquid)	$\overline{\nabla T}$	$K\ m^{-1}$	3708.2
Low thermal gradient surface tension	σ_0	$N\ m^{-1}$	-1.209
Low thermal gradient surface energy	γ_0	$J\ m^{-2}$	0.168
Surface stress	s	$N\ m^{-1}$	-0.631105

Table 1. Continued...

Properties	Symbol	Units	Al-3wt%Cu-0.5wt%Mg-0.2wt%Fe
Surface tension	σ_{SL}	$N\ m^{-1}$	-1.84011
Surface energy	γ_{SL}	$J\ m^{-2}$	0.915939
1 st Order Gibbs-Thomson	$\Gamma_{Het}^{1st-Order}$	$K\ m$	4.17091×10^{-7}
2 nd Order Gibbs-Thomson	$\Gamma_{Het}^{2nd-Order}$	$K\ m$	1.09735×10^{-6}
Dendrite tip growth rate horizontal	v_L	$mm\ s^{-1}$	$2.40P^{-0.305}$
Local solidification time horizontal	t_{SL}	s	$1.60P^{1.00}$

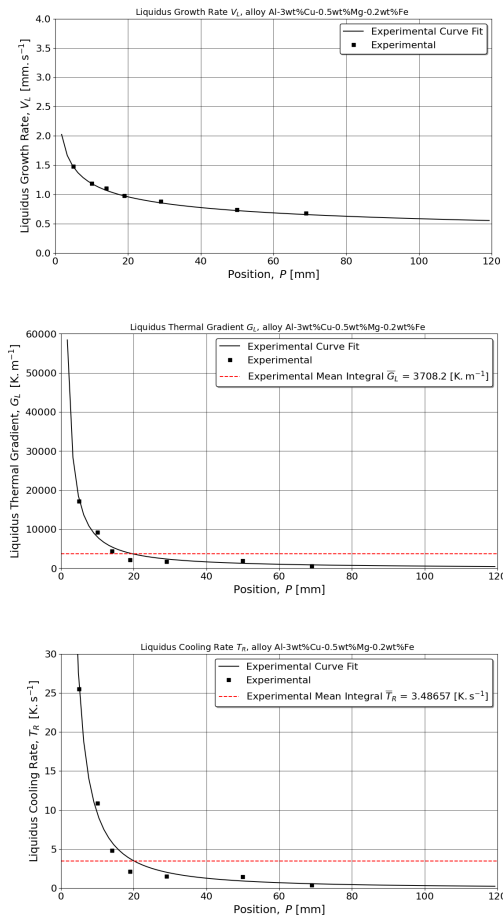


Figure 2. Experimental data from transient solidification measurements yielded (A) dendrite tip growth rates, (B) liquidus thermal gradients, and (C) dendrite tip cooling rates.

$\Gamma^{1st-Order}$ and $\Gamma^{2nd-Order}$ are provided as a function of increasing thermal gradient ∇T in Table 2.

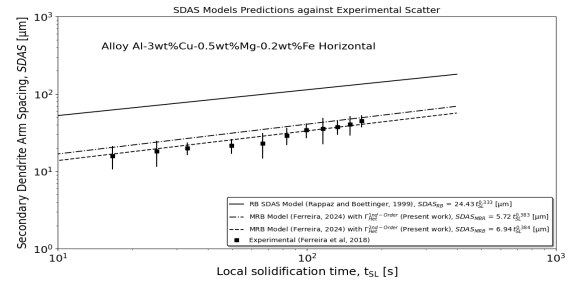


Figure 3. Comparison between secondary arm-spacing models and experimental scatter.

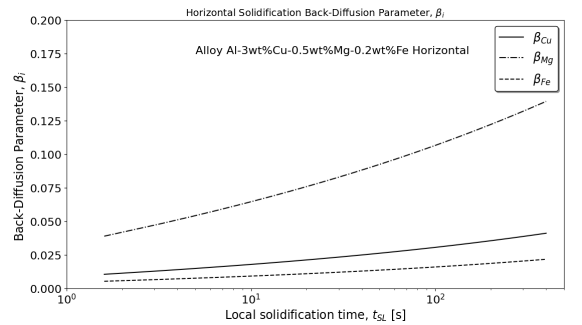

 Figure 4. Theoretical prediction of local back-diffusion parameters β_i .

Figure 2 shows the experimental thermal gradient against the position and the mean thermal gradient applied to determine the non-equilibrium nucleation properties for the FCC_A1 phase in Table 2.

In Figure 3, the SDAS calculations are carried out for the Al-3wt%Cu-0.5wt%Mg-0.2wt%Fe alloy and plotted against the experimental data found in³⁰ as a necessary step for evaluating Equations 11-17 related to the microsegregation parameter β_i for each solute as presented in Figure 4. This step is necessary to accurately compute microsegregation patterns; Equation 7, as β_i falls between the limits defined by the Scheil and Lever Rule models – representing finite diffusion rather than 0 or 1.

Table 2. Calculated nucleation variables as a function of thermal gradient for the FCC_A1 phase of the Al-3wt%Cu-0.5wt%Mg-0.02wt%Fe alloy²⁵.

∇T	ΔT	$r_{C.Hom}$	$r_{C.Het}$	θ	$\frac{\partial f(\theta)}{\partial r}$	σ_{SL}	Σ	γ_{SL}	$\frac{\partial \gamma_{SL}}{\partial r}$	Γ^{1st}	Γ^{2nd}
$K.m^{-1}$	K	m	m	m	$1/m$	$1/m$	$N.m^{-1}$	$J.m^{-2}$	$J.m^{-2}$	$m.K$	$m.K$
443.579	0.0698834	6.2685E-06	6.26849E-06	3.06676	-15.9999	-15.9999	-0.0121204	0.182823	-440205	2.56276E-08	2.19033E-07
1030.02	0.151674	0.000005859	5.85898E-06	3.05653	-15.9998	-15.9998	-0.175477	0.370875	-479403	9.28244E-08	4.4433E-07
2004.31	0.271338	5.3865E-06	5.38648E-06	3.0506	-15.9998	-15.9998	-0.378791	0.609973	-534742	2.17423E-07	7.30784E-07
3016.2	0.382059	0.00000504	5.03998E-06	3.04819	-15.9998	-15.9998	-0.539564	0.803627	-584554	3.39841E-07	9.62793E-07
3708.2	0.452259	4.85277E-06	4.85274E-06	3.04734	-15.9998	-15.9998	-0.631105	0.915939	-615670	4.17091E-07	1.09735E-06
4112.97	0.49168	4.7565E-06	4.75648E-06	3.047	-15.9998	-15.9998	-0.679553	0.976031	-633025	4.5992E-07	1.16934E-06
5022.88	0.576596	4.5675E-06	4.56748E-06	3.04652	-15.9998	-15.9998	-0.777594	1.09912	-670186	5.50379E-07	1.31681E-06
6115.05	0.672923	4.3785E-06	4.37848E-06	3.04625	-15.9998	-15.9998	-0.879779	1.22966	-712067	6.49625E-07	1.4732E-06
7196.24	0.763415	0.000004221	4.22098E-06	3.04617	-15.9998	-15.9998	-0.968361	1.34484	-751240	7.39428E-07	1.6112E-06
8196.26	0.843547	0.000004095	4.09498E-06	3.0462	-15.9998	-15.9998	-1.04164	1.44164	-785844	8.16194E-07	1.72717E-06
9038.27	0.90874	4.0005E-06	4.00048E-06	3.04626	-15.9998	-15.9998	-1.09809	1.51722	-813967	8.76795E-07	1.81772E-06

4. Conclusions

From the results obtained, the following conclusions:

A novel scheme is proposed for determining the solute redistribution coefficient β_i based on the $SDAS/2$ Fourier number α_i calculated from the nucleation formulation recently derived by Ferreira et al.^{20,24}.

The nucleation/growth radius r_C is used to demonstrate this proposal in which the local Fourier number α_i is theoretically determined from the nucleation model in respect to the thermal field tensor Γ .

The present analysis may strongly contribute to the understanding of how to manage solidification operational parameters aiming at designing of the LHTES reservoirs.

This work will serve as a base calculation to determine non-equilibrium Gibbs Free-energy and other thermodynamic quantities, based on the dependence of surface energy throughout Computational and Statistical Thermodynamics, by considering calorimetric and spectral analysis.

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Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.